Parth Natekar

Department of Engineering Design, IIT Madras

C-1101, Beverly Hills
Pancard Club Road, Baner
Pune, Maharashtra, India 411045
Github | parthnatekar26@gmail.com
Website | LinkedIn | +91-99-609-50978

Education

INSTITUTION & DEGREE	CGPA	YEAR
Indian Institute of Technology Madras *Bachelor of Technology in Engineering Design,* *Master of Technology in Biomedical Design*	**8.89/10.00**	2020
Fergusson College *Higher Secondary School (Major: Electronics)*	92.4 %	2015
Vikhe Patil Memorial School *Secondary School (CBSE)*	97.0 %	2013

Publications

Parth Natekar, Manik Sharma. "Representation Based Complexity Measured for Predicting Generalization in Deep Learning." *Winning Solution of the NeurIPS 2020 competition on Predicting Generalization in Deep Learning*. (Preprint). (A more detailed version will be published as part of the NeurIPS 2020 post-proceedings with runner-up teams, under completion).

Parth Natekar, Avinash Kori, and Ganapathy Krishnamurthi. "Demystifying Brain Tumour Segmentation Networks: Interpretability and Uncertainty Analysis."
Frontiers in Computational Neuroscience 14 (2020): 6 (Link).

Avinash Kori, **Parth Natekar**, Balaji Srinivasan, and Ganapathy Krishnamurthi.
"Abstracting Deep Neural Networks into Concept Graphs for Concept Level Interpretability."
arXiv preprint arXiv:2008.06457 (2020). (Link).

Research & Professional Experience

Anheuser-Busch InBev, Bangalore

Data Scientist | Machine Learning for Compliance Analytics *Aug '20 - Present*

- Working on building machine learning models and developing hypotheses to flag anomalous and risky transactions for legal and financial compliance.

Medical Imaging and Reconstruction Lab, IIT Madras

Research Associate | Interpretable Deep Learning for Medical Image Analysis

(Guide: Dr. Ganapathy Krishnamurthi, Engineering Design, IIT Madras) June '19 - June '20

- Conceptualized a pipeline for interpretability and uncertainty estimation of deep learning models which perform various image processing tasks in the medical domain, such as tumor segmentation.
- This involved developing a new method for concept-level interpretability as well as implementing techniques like Network Dissection and Bayesian Dropout on segmentation and classification models.
- Implemented the pipeline on benchmark brain tumor segmentation datasets to understand information flow and representation in internal layers and to quantify uncertainty of model predictions.
- Developed a python package for direct implementation of the proposed interpretability pipeline on deep neural networks for any relevant biomedical dataset.

Honeywell Technology Solutions, Bangalore

Research Intern | Graphical Modelling of Neural Networks for Causal Inference

(Mentor: Satyanarayan Kar, Honeywell Aerospace, India) *January '19 - June '19*

- Devised a probabilistic graphical model based algorithm for Causal Inference on a deep neural network, to provide human-understandable explanations of what the network actually learns.
- Implemented the algorithm on benchmark datasets and successfully delineated variables in the model which were causal for a particular prediction compared to those with spurious correlations.
- Developed a deep learning pipeline for time-series analysis of flight data in order to predict anomalous flight landings. Achieved 86% accuracy in predicting instability 2 minutes prior to landing.
- Implemented an algorithm for object detection and text extraction from aircraft documents.

Convergence Design Lab, Purdue University, USA

Research Intern | Wearables for Human-Computer Interaction

(Guide: Dr. Karthik Ramani, Mechanical Engineering, Purdue University) *May '18-July '18*

- Completed the entire design cycle – ideation, detail design, fabrication and prototyping, of IOT based flexible, wearable electronic devices for interaction with Mixed Reality environments.
- Devised a novel method to design flexible electronic circuits on various fabric substrates. Conducted material and sensor tests to integrate tactile sensing using carbon based piesoresistive sensors.
- Developed multiple ergonomic, working prototypes. Set up networking capabilities via UDP/TCP protocols and tested the wearables in Mixed Reality and IoT environments.
- Created applications in Unity and C# to validate the efficiency of using the prototyped wearables.
- Reduced task time by 44% and error rate by 74% vis-à-vis traditional interaction methods.

CERN (European Organization for Nuclear Research), Geneva

Engineering Intern | Testing Apparatus for CMS cooling system

(Mentor: Antti Onnela, EP-DT, CERN, Geneva) *May '17 - July '17*

- Selected to be part of a 3-member team from IIT Madras to work on the Large Hadron Collider.
- Designed an adiabatic testing apparatus for the evaporative CO_2 based cooling system of the CMS Detector using concepts from thermal, mechanical and electrical design.
- Set up sensing and feedback system for monitoring of performance of the cooling system
- Created a software interface in LabView to enable easy user interaction.

Awards & Achievements

- **Winner, NeurIPS competition on Predicting Generalization in Deep Learning**, 2020
- **PURE fellow**: Awarded PURE Scholarship for a two-month Research Internship at Purdue University, Indiana, having been selected in the top 14 among 800 students from IIT Madras.
- **CBSE Class X School Topper**: Came first in school with percentage of 97.00%. Awards for highest marks in Math (100/100) and Science (99/100).
- One of 15 students selected from 5000 by Infosys for their **'Spark – Catch Them Young' Program**.

Related Course Work

Computation & Machine Learning

- Machine Learning for Engineering Applications
- Probabilistic Graphical Modelling
- Computational Systems Biology
- Computational Methods in Design
- Linear Algebra for Engineers
- Introduction to Computation and Visualization
- Data Structures and Algorithms in Python

Biomedical Design

- Medical Image Analysis
- Design of Diagnostic and Monitoring Systems
- Design of Implantable and Surgical Devices
- Medical Equipment Dissection Lab
- Human Anatomy, Physiology & Biomechanics
- Biomaterials Engineering

Mechanical/Electrical Engineering

- Design of Mechanical Systems (I & II)
- Design of Electronic Systems (I & II)
- Digital Signal Processing
- Engineering Mechanics
- Design of Thermal and Fluid Systems
- Electromagnetic Compatibility for Design
- Bio-Micro-Electrical-Mechanical Systems

Design Methodology & Art

- Functional and Conceptual Design
- Human Factors in Design
- Geometric Modelling and CAD
- Creative Design
- Design for Form and Aesthetics
- Computational Methods in Design

Skills

Programming: Python, C, C++, C#, JavaScript, Unity, Machine Learning (Keras, Tensorflow, Pytorch), Arduino, Raspberry Pi, LaTeX, SQL
Languages: Competent French, Limited Working German.

Relevant Course Projects

Computational Modelling of Pathogen Interactomes

January to May 2018 | Supervisor : Dr. Karthik Raman

- Delineated drug-resistance pathways in the pathogen Staph Aureus through systems level computational modelling of the protein-protein interaction network of the organism.
- Devised a modified network-based shortest path algorithm by integrating node centrality measures with chemical protein interaction scores between drug molecules and proteins.
- Achieved 68.5% prediction accuracy on already known resistance pathways, validating the viability of our algorithm for exploring drug resistance pathways in pathogens.

Colour Sensing Robotic Manipulator

January to May 2017 | Supervisor : Dr. N. J. Vasa

- Designed and implemented a 6-dof color detecting robotic manipulator with a magnetic tooltip arm.
- Involved electronic circuit design with the use of a microcontroller to control the robot.
- Coded in C/C++ using the Arduino IDE.

Ultrasound Nerve Image Segmentation

Jan to May 2017 | Supervisor : Dr. Ganapathy Krishnamurthi

- A deep learning-based project for semantic segmentation of ultrasound nerve cell images.
- Used a fully-convolutional neural-network implemented in Python using Keras and Tensorflow.
- Obtained 71.2% Dice Score on test dataset – on par with then state-of-the-art techniques

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Teaching and Voluntary Experience

Teaching Assistant - Design of Electronic Systems Lab, IIT Madras

Professor :Dr. Tuhin Subhra Santra *Aug ’19 - Dec ’19*

- Selected as Graduate Teaching Assistant for the Design of Electronic Systems Lab (Fall 2019).
- Managed and guided a group of 60+ students in electronic circuit design principles.

Teaching Assistant - Medical Image Analysis, IIT Madras

Professor :Dr. Ganapathy Krishnamurthi *Aug ’19 - Dec ’19*

- Conducted tutorial sessions to teach students the programming aspects of Medical Image Analysis.

Deputy Head, International Relations Cell, IIT Madras *July ’17 - July ’18*

- Led and organized a team of 20+ people, managed a group of 80+ international exchange students.
- Helped organize various events, including ‘International Day’, which catered to 2000+ people.

Volunteering

- Worked with an organization called *Lata Mahajan Foundation* on their project *EducationLinked*, which is aimed at overall development of tribal and underprivileged children.
- Worked with *iVolunteer*, Bangalore to help organize health camps in orphanages.

Extra Curricular Activities

Institute Football Team *Aug ’16 - Dec ’17*

- One of only two freshers to be a part of the Institute Football Team.
- Captain of School Football Team. Participated in CBSE West Zone Tournament in 2011 and 2012

Literary and Debate

- Secured the 3rd position for Ganga Hostel, IIT Madras in the Lit-Soc inter-hostel debate competition.
- Part of organizing committee of IIT Madras Model United Nations 2017.

Art and Design

- Selected Impressionist paintings uploaded on Behance.